A bank run model for the twentieth century

Who runs, and how a run spreads: an agent-based test of the theory

Walter Stover · George Mason University · 26 October 2026

The puzzle: who runs, and how a run spreads

Diamond–Dybvig

A run is an equilibrium. The theory says runs can happen; it says nothing about who runs first or how a run spreads.

Chapter 1: theory

Depositors differ in how much they weigh their own information against public news and what others do. That difference predicts who moves first, and in which direction, given the news.

Chapter 2: Celsius

Those differences show up in real withdrawals. But one platform and one shock cannot vary the mix of depositors or the information they received.

Chapter 3, this talk: a population of interacting depositors in which the mix of types, the public signal, deposit concentration and the bank's contract can each be varied. Does the theory's mechanism survive there?

D — share of individualists who withdraw minus share of collectivists, among depositors the shock did not hit

P(fail) — share of runs that end with the bank failing

Two mechanisms the literature has used

1

Who runs, and when, has been answered two ways: by luck and by position.

Dispersion. Goldstein–Pauzner give every depositor a noisy draw of the same signal. The draw breaks common knowledge and selects the equilibrium. Who runs first is whoever drew low.

Delivery. Networks and media decide who hears what, from whom, when (Kelly–Ó Gráda, Iyer–Puri, Cookson et al.). Who runs first is whoever sits nearest the source.

Both are fruitful. Neither says anything about the depositor: the same Bayesian, with the same prior, sits at every node.

With two signals, a third question appears

2

Which signal does a depositor act on, and in what proportion?

Latent in the canonical models. Diamond–Dybvig has no signal; Morris–Shin and Goldstein–Pauzner have one. The question cannot arise.

Askable once a public signal is added. Morris–Shin (2002), Hellwig (2002), Angeletos–Werning (2006) put a public signal next to the private one.

The question a panic poses. A depositor in a run holds two things that disagree: what they see and what they are told.

How the theory answers it: the Bayesian precision ratio

3

Weight on the public signal = its precision relative to the private one. The same for everyone, and vanishing.

The same for everyone. Two depositors with the same signals form the same belief. There is no ‘who’.

Vanishing. As private information sharpens, the weight on the public line goes to zero. The public signal matters only while depositors know little.

What it predicts about a panic. Any sorting between depositors is noise, and official communication stops mattering just when depositors are paying most attention.

Panics do not look like that

4

Panics are open disagreements between the public line and private evidence, and depositors split on which to trust.

Celsius, 2022. Official posts insisting the platform was sound, while CEL fell and withdrawals accelerated.

Silicon Valley Bank, 2023. A call asking clients to stay calm, against a Twitter feed saying the opposite.

Northern Rock, 2007. A Bank of England support announcement, against queues outside the branches.

In each case the interesting fact is who moved on which signal.

Cultural economics: a source of stable weights

5

Culture supplies a weight that differs across depositors, is set before the run, and does not move with the information environment.

Transmitted, not chosen. Cultural values pass largely unchanged from parents to children (Guiso, Sapienza, Zingales; Bazzi et al.). The weight is a disposition, not a calculation.

Individualism is the right dimension. Self-reliance against deference to the group (Hofstede) is the private-against-social weighting the model already contains (Gorodnichenko–Roland).

Measurable per depositor. Given names carry it (Twenge et al.), so the weight can be observed, not assumed.

What this dissertation does with it

6

The weight is the depositor's psychological characteristic; culture is the population-level source.

Chapter 1 formalises it. A stale public signal and heterogeneous dispositional weights inside Goldstein–Pauzner. Prediction: who runs first reverses with the sign of the public error.

Chapter 2 observes it once. Celsius: individualism predicts direction and timing within platform and within ancestral origin, and which public signals depositors act on.

Chapter 3 varies it. The mix of types, the gap between them, the staleness of the news, and the bank's contract, in a population of interacting depositors.

Why the third chapter is a simulation

7

Chapter 1 relaxed one assumption. The agent-based model relaxes the rest at once, and lets the three rungs interact.

Chapter 1 relaxed one thing. The precision ratio became a stable, heterogeneous weight on a stale public signal. Everything else stayed: a continuum of depositors, one simultaneous move, no network.

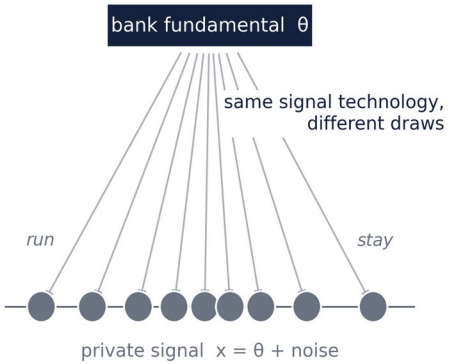
The model relaxes the rest at once. Dispersion (a private draw per depositor), delivery (a network, neighbours' withdrawals arriving tick by tick, public news absent, accurate or stale), interpretation (λ, formed in a cultural warm-up), plus deposit sizes, risk preferences and the contract.

An analytical model cannot carry all of that. Each addition breaks the threshold-equilibrium machinery; together they are beyond closed form. The simulation is what lets the rungs be varied and measured separately.

The ladder: dispersed, delivered, interpreted

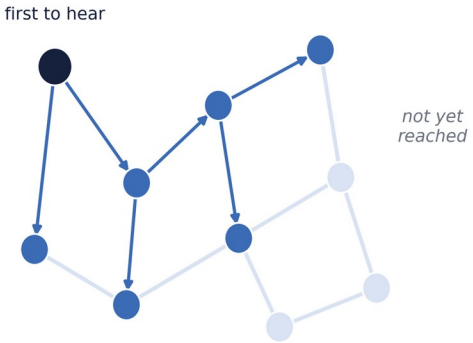
1 Dispersed

one fundamental, one noisy draw each



2 Delivered

who hears it, from whom, when

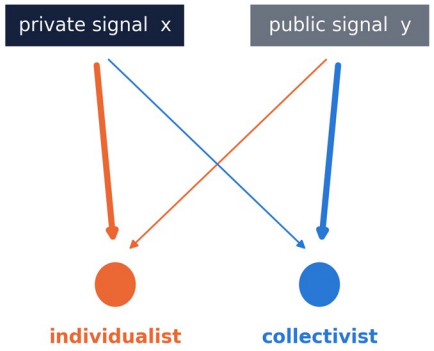


networks and media: Kelly–Ó Gráda, Iyer–Puri, Cookson et al.

3 Interpreted

the same two signals, different weights

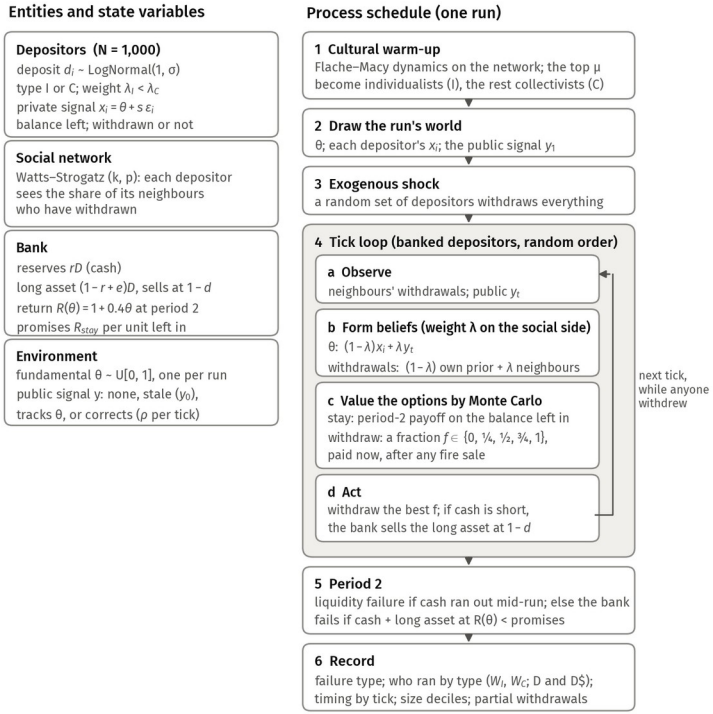
culture sets the weight



leans on what it sees leans on the public line

Rungs 1 and 2 are the literature. **Rung 3 is this dissertation:** the only one that orders depositors by a trait, and reverses when the public line is too optimistic or too pessimistic.

The model in one picture



1,000
depositors on a small-world network, one bank

1%
cap on any one deposit; sizes log-normal (σ 2.5) below it

CARA 0.1
risk aversion; partial withdrawals allowed

250
runs per cell: 25 initialisations × 10 draws

- One bank that sells its long asset at a discount when cash runs short (a fire-sale bank)
- Each tick a depositor sees neighbours' withdrawals and the public news, blends them with a private signal, and withdraws if that pays
- A public signal can be absent, accurate, or outdated

Individualists and collectivists: who listens to whom

Individualists

- Trust their own information about the bank
- Put little weight on what neighbours do or on public news
- Weight on the social side: $\lambda = 0.1$ to 0.3
- Act on what they see themselves

Collectivists

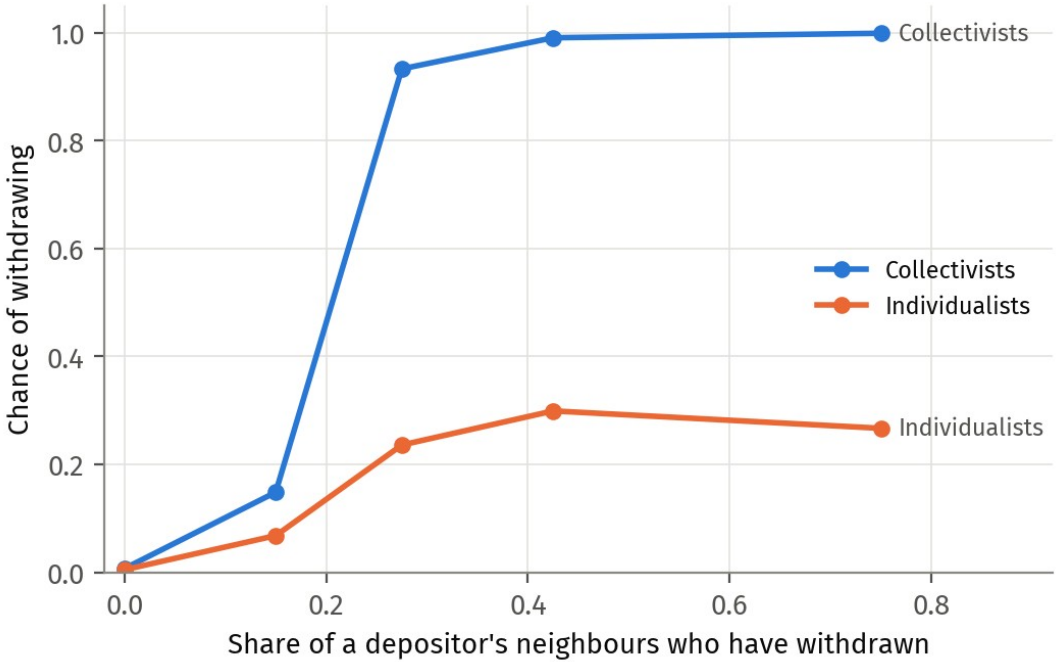
- Watch what neighbours do and what the public news says
- Put little weight on their own information
- Weight on the social side: $\lambda = 0.5$ to 0.9
- Move when others move

Three pairs (λ individualist, λ collectivist) in the figures

(0.1, 0.9) wide gap **(0.2, 0.8)** medium **(0.3, 0.5)** narrow

The share of individualists μ is set by a cultural warm-up on the network. **Orange = individualists**, **blue = collectivists** throughout.

Collectivists herd



Chance of withdrawing, given the share of a depositor's neighbours who have already withdrawn. λ pairs pooled; reference bank.

0.15 to 0.93

collectivists, once 20–35% of their neighbours are out

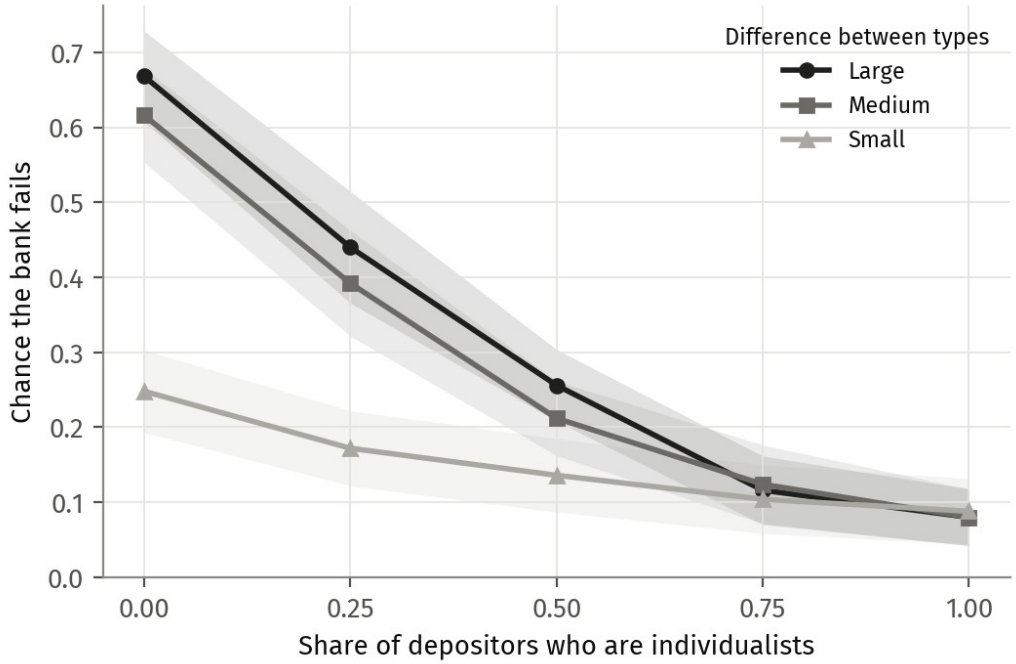
0.25 to 0.30

where individualists level off, whatever their neighbours do

+0.58

difference in slopes (collectivists 0.80, individualists 0.22), t = 16.7

More individualists, fewer failures



Chance the bank fails, by share of individualists μ, one line per λ pair. Includes the pure populations (μ 0 and 1).

0.67 to 0.08

P(fail) from no individualists to all, wide λ gap

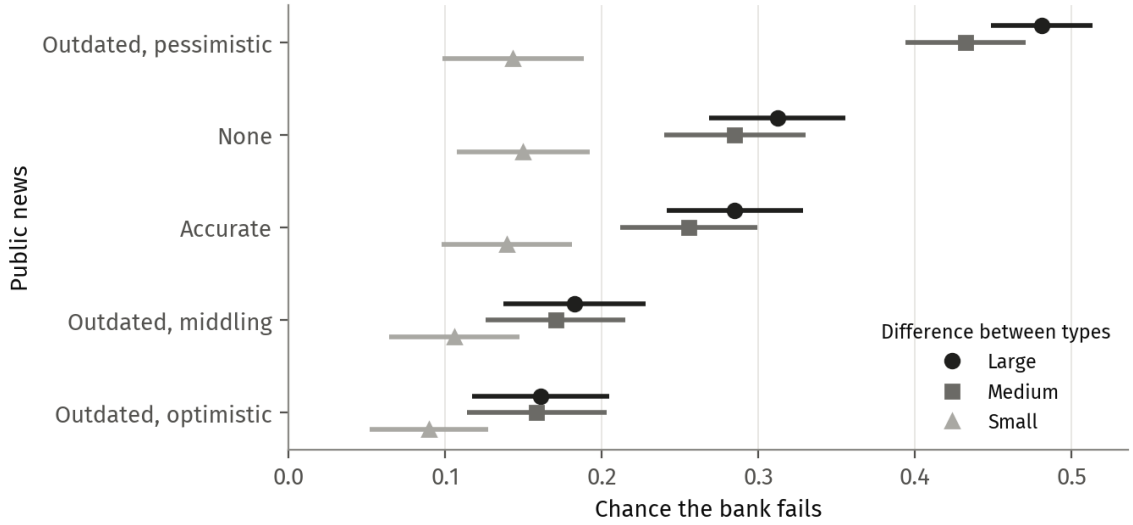
0.25 to 0.09

the same, narrow λ gap: it matters most when collectivists dominate

Monotone

No interior peak at any mix. The lines converge once individualists are 75% or more.

Outdated news cuts both ways



Chance the bank fails, by the kind of public news, dot and CI per λ pair. All mixes μ. Numbers at right: wide λ gap.

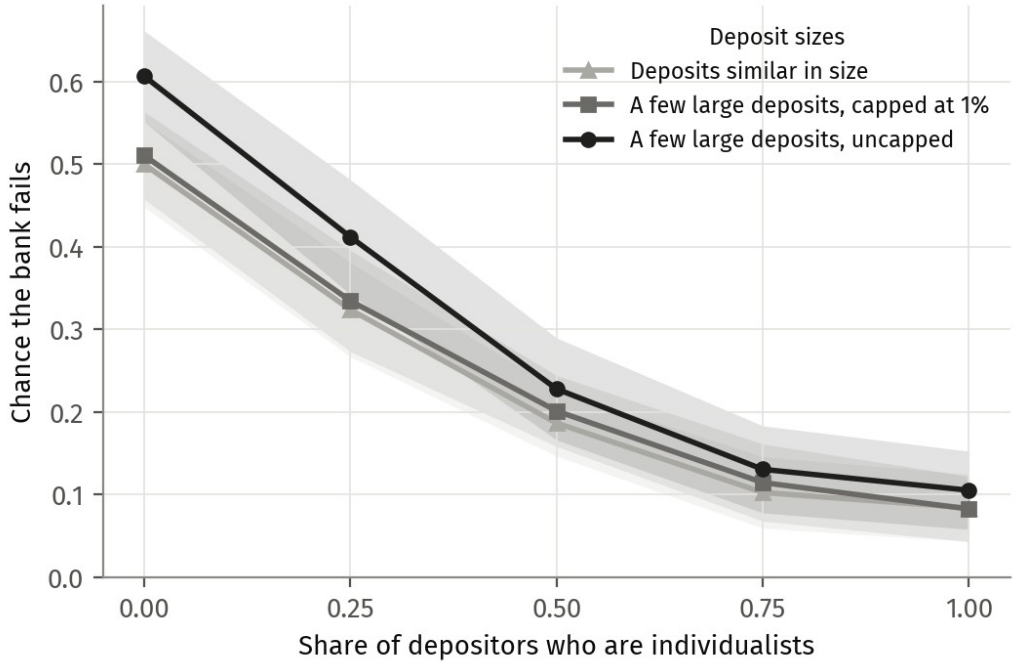
0.48
outdated, pessimistic news

0.31
no public news

0.29
accurate news

0.16
outdated, optimistic news

Capping the largest deposits



Chance the bank fails by share of individualists, for three deposit-size distributions. λ pairs pooled.

0.61 to 0.11

uncapped: a few very large deposits (σ 2.5)

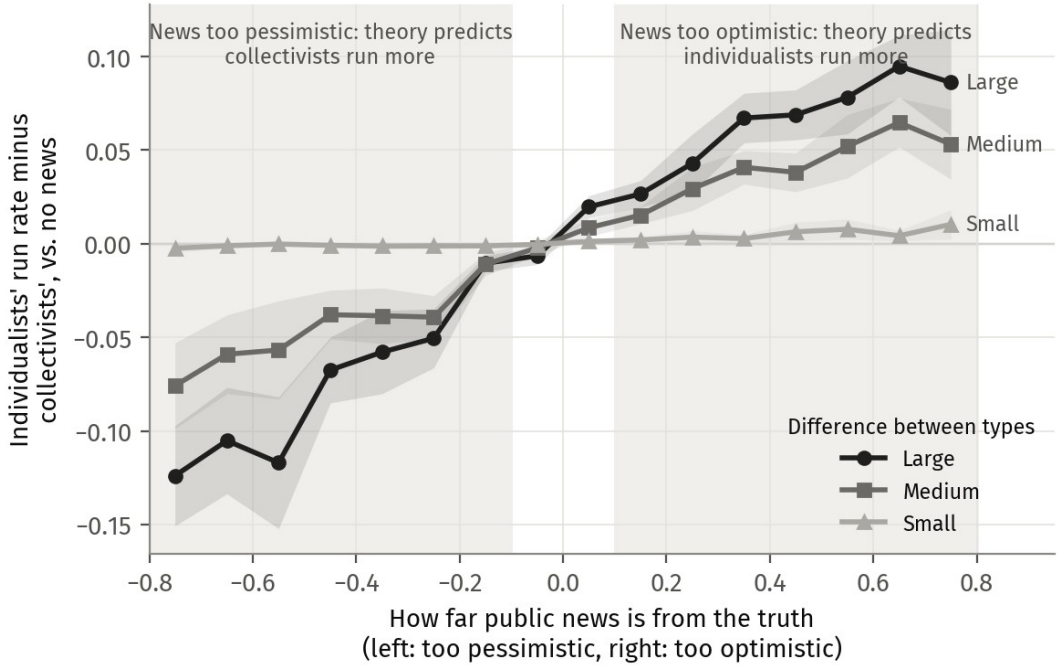
0.51 to 0.08

capped: no deposit above 1% of the total (the reference bank)

0.50 to 0.08

similar sizes (σ 1). The μ gradient is the same in all three.

Does the model hold up?



D relative to no news, against how far the public news sits from the truth. Shaded: where Chapter 1 predicts the sign.

Pre-registered tests

G1′ optimistic news: individualists run more +0.033

G4′ pessimistic news: collectivists run more −0.031

G2a′ the effect grows with the λ gap +0.093

G3 sorting with no news: counts against (one pair wrong-signed)

3 of 4

confirmatory tests support the theory, on all three versions of the bank

Takeaways and next steps

1	**Sorting follows the news**	Optimistic public news makes individualists run relatively more; pessimistic news, collectivists. 3 of 4 pre-registered tests pass, in all three bank versions.
2	**Collectivists herd**	Their chance of withdrawing jumps from 0.15 to 0.93 once 20–35% of neighbours are out; individualists level off near 0.3.
3	**More individualists, fewer failures**	P(fail) falls from 0.67 to 0.08 as the individualist share goes from 0 to 1. Monotone; no interior peak.
4	**Stale news cuts both ways**	A stale optimistic signal halves failure, a stale pessimistic one raises it by half. Accurate news barely moves it.
5	**Capping the largest deposits**	A 1% cap brings failure down to the no-concentration bank's level; the culture gradient is unchanged.
6	**Not yet matched**	How Celsius's largest depositors exit: one-shot share 0.35 there, 0.93 or more in the model.

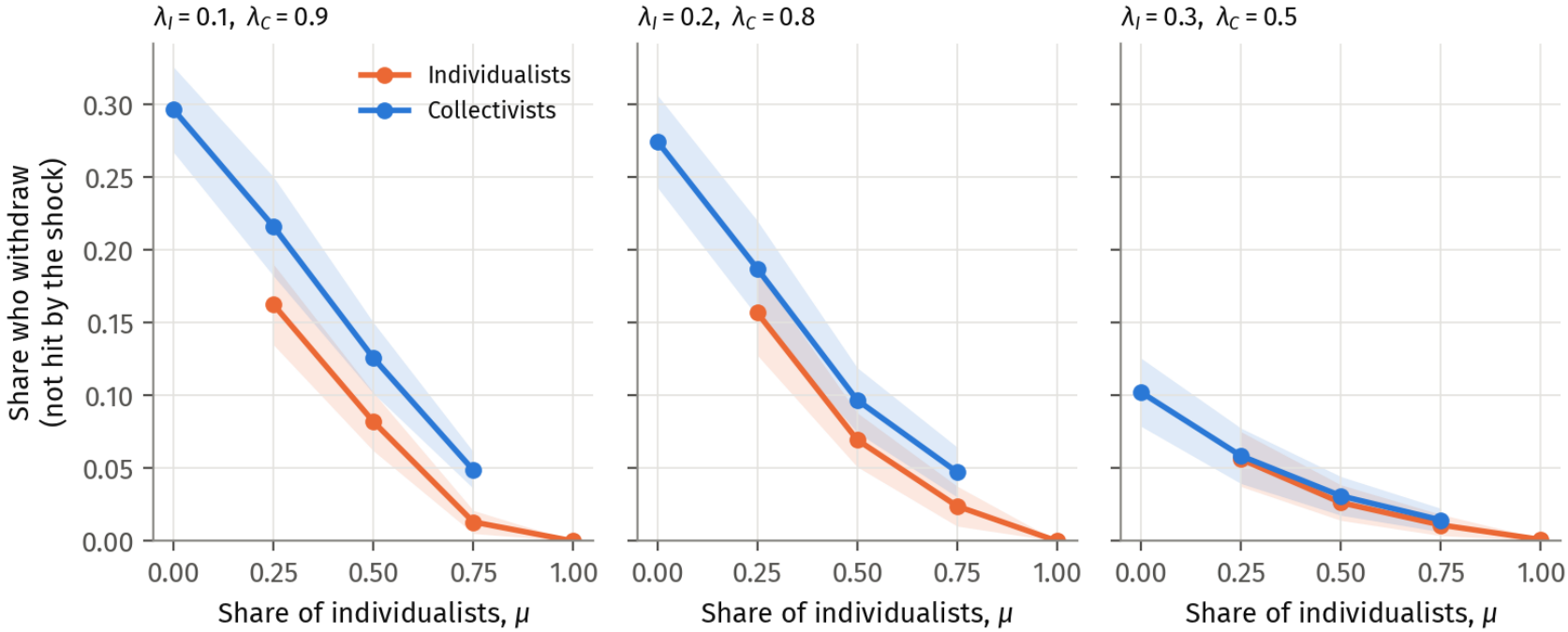
Next steps

- Super-individuals, to match how the largest depositors exit (Chapter 2 docking)
- Deposit insurance and the reserve ratio on the fire-sale bank
- Network topology and learning
- Full sweep: stale public signal plus no-signal baseline, CARA 0.1, σ reported

Backup

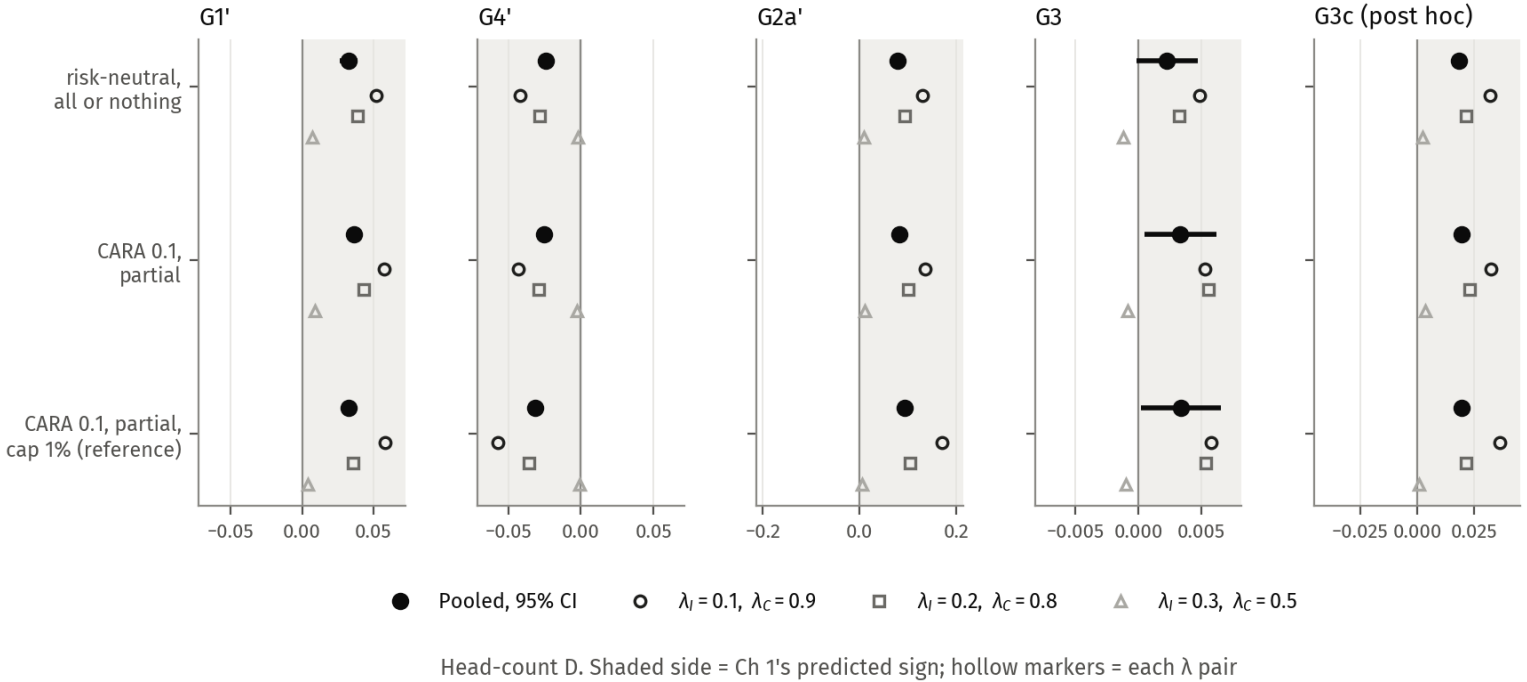
Technical versions of the exhibits, the pre-registered tests, channels, and the Celsius docking gap.

Who withdraws, by type and mix



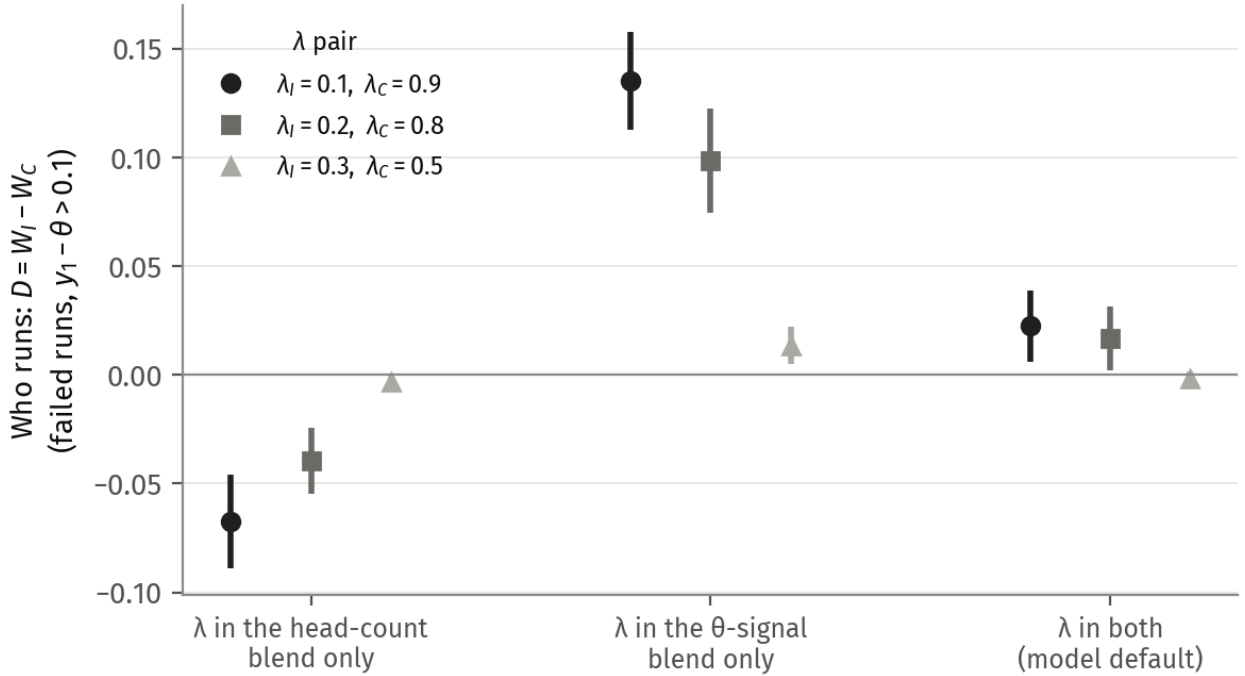
Share who withdraw by individualist share μ, one panel per λ pair; orange individualists, blue collectivists, with CIs.

The pre-registered tests, three versions of the bank



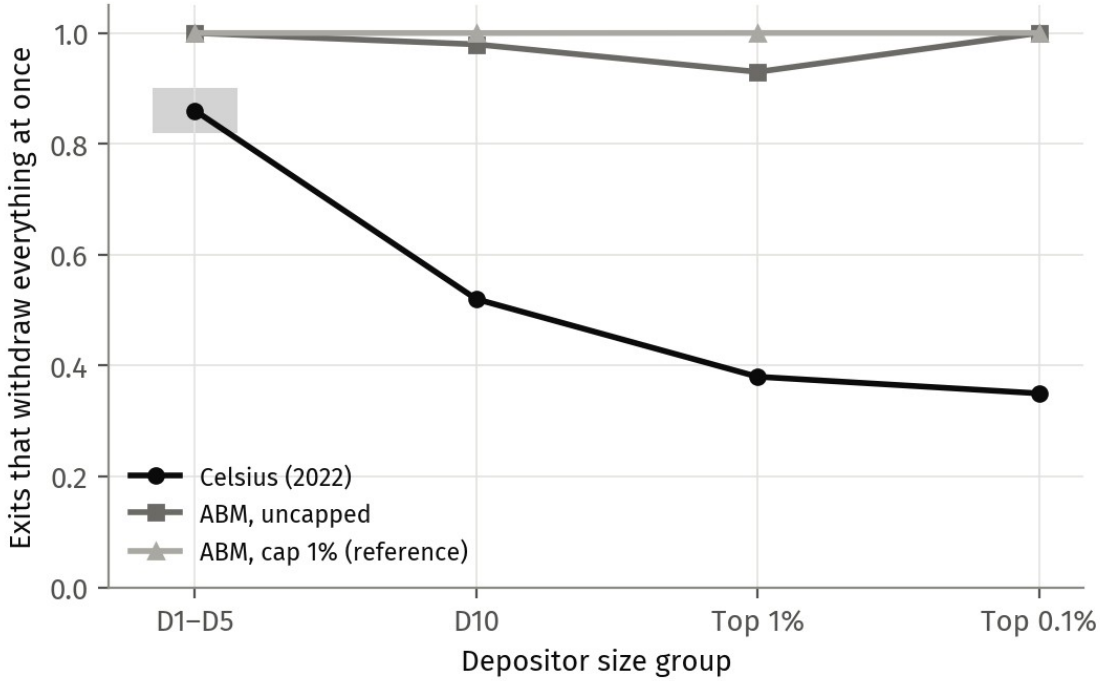
G1′, G4′, G2a′, G3 and G3c (post hoc). Rows: risk-neutral all-or-nothing, CARA partial, capped reference bank. Filled dot: pooled; hollow: each λ pair. Shaded: Chapter 1's predicted sign.

Which channel carries culture



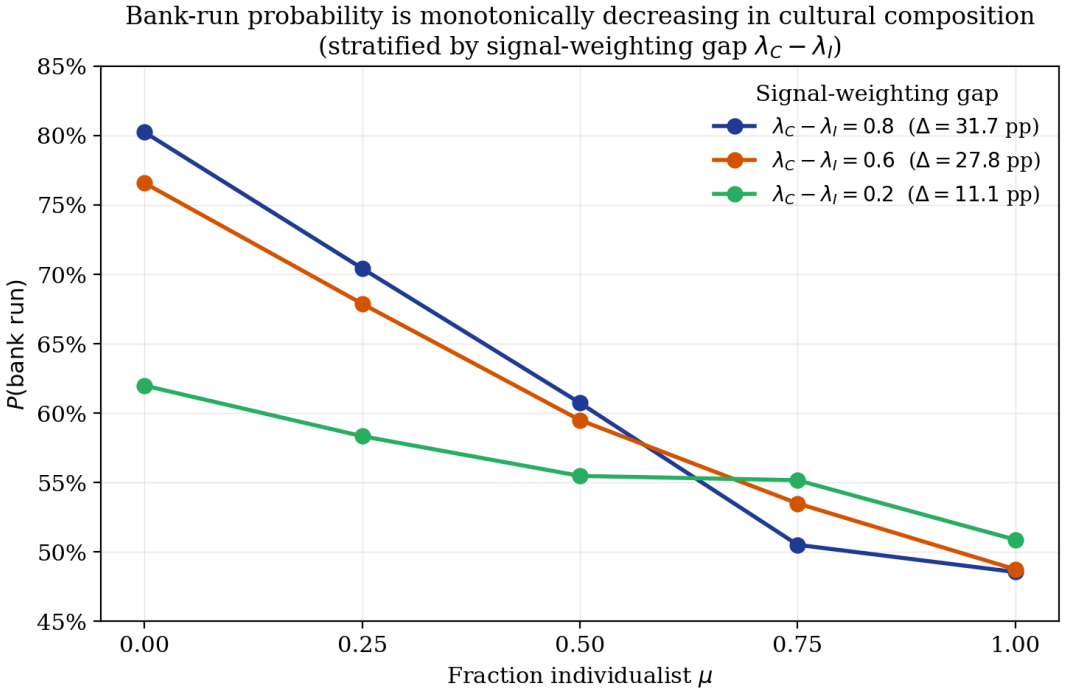
D in failed runs with optimistic news, when λ weights only the neighbour blend, only the news blend, or both (the model). θ-blend only +0.14; head-count only −0.07; both +0.02.

Matching Celsius: how large depositors exit



Share of exits that withdraw everything at once, by depositor size. Black: Celsius (band for D1–D5); greys: the model, uncapped and capped.

Earlier specification (vault bank): failure by mix



Vault bank, earlier specification (chapter Figure 1): probability of a run by share of individualists, three λ gaps. Also monotone.